



Before you commit a large experiment, find out whether it can teach you enough.

EvoForecast is a reproducible digital wind tunnel. It asks a narrower and more useful question than whether evolution is predictable. Can a proposed observation programme tell prospective forecasts apart before a costly physical programme is built?

5,600/5,600

latest blind synthetic SLiM
5.2 runs completed
successfully

0/60

latest forward-looking
cells that reached the
indicative 50 percent bar

0/120

latest experiment plans
classified robustly best

Claim boundary. These results demonstrate synthetic software and experiment-design evidence. Forecast skill in real organisms and biological qualification await empirical testing. Independent review precedes any operational, partnership, or funding claim.

The choice that sets up every other choice.

Species, generations, populations, starting diversity, conditions, assays, and automation form one coupled design. The wind tunnel tests which combinations resolve a forecast and which add observations while leaving designs indistinguishable.

Claims EF-001 to EF-006 · exact source hashes in audit/source_manifest.json

An auditable chain from parameters to a scored forecast.

REGISTERED PARAMETERS	SLiM 5.2 WORLD	LATENT STATE
OBSERVATION MODEL	MODEL FORECAST	IMMUTABLE DEPOSIT
INDEPENDENT REVEAL	PROPER SCORE	DESIGN / ABSTAIN

Engine direction. SLiM 5.2 is the registered trajectory engine for this round. The challenge harness is engine-agnostic, so other evolutionary simulators can enter future rounds and are under evaluation. Ecological network dynamics in a population context are a direction for later work.

What the latest challenge says

The genome-plus-phenome track, combining genome data with measured traits, improved its mean continuous ranked probability score, or CRPS, by 47.5 percent over the pedigree baseline, aggregated across 35 registered contexts. CRPS measures forecast accuracy. Its ninety percent prediction range held the truth eight times in ten. Zero forward-looking cells reached the combined bar. Average accuracy, calibration, population-level identifiability, and transfer are separate tests.

Interpretation. This is informative synthetic design evidence. Biological forecast skill awaits an empirical round.

Why controls matter

A corrected favorable marker control produced a 17.0 percent genome-versus-pedigree gain. A matched adverse control produced minus 9.2 percent. The opposite signs show the workflow can expose failure and avoid manufacturing a genomic advantage.

Supported claim. The software control behaves differently under registered favorable and adverse conditions.

The signature instruments

The build-the-experiment view returns registered gain, coverage, uncertainty, and combined-bar probability for a chosen track, population count, and horizon. The measurement-plan frontier places forecast value against modeled resource use for named packages. The trajectory view exposes all 32 registered histories in one Study E persistence cell, including exact daily population counts and founder composition. Every browser control selects a frozen row.

Software proof is in hand. Biological qualification is the next gate.

1 · Software proof	SUPPORTED · SYNTHETIC
2 · Indicative design evidence	CONDITIONAL
3 · Empirical qualification	BLOCKED
4 · Operational forecasting	FUTURE

One trajectory cell, fully exposed

Each adaptive_E07 vessel begins with 80 age-synchronized grazers, ten from each of eight immutable founders. All 32 retained grazers through day 90. Standing selection still produced different founder and population histories. This is a registered synthetic persistence diagnostic, and its value lies in the distribution that practitioners would need to forecast.

What remains blocked

Empirical qualification requires exact material identity, accepted starting measurements, real assay performance, material custody, independent prospective reveal, uncertainty analysis, safety authority, and external reproduction. Diploid sexual genetics and ecological network dynamics remain future directions.

Run one small round together.

Bring one candidate keystone organism, one contained stressor, one feasible budget stated in USD, and one outcome sealed until scoring. We rehearse the design, then run the smallest real round that can estimate calibration and forecast accuracy under an independent reveal. A later contained round can test whether a targeted intervention improves resilience after the forecast layer earns trust.

EvoForecast turns prediction into a testable programme decision. Measure what can discriminate, seal the outcome, score the forecast, and scale only after the evidence earns it.

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